

Towards improving Disfluency Detection from Speech using Shifted Delta Cepstral Coefficients

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ABSTRACT

In this paper, we investigate the use of shifted delta cepstral (SDC) coefficients for detecting disfluencies in two types of speech - stuttered speech and spontaneous lecture-mode speech. SDC features capture temporal variations in the speech signal effectively across several frames. The UCLASS stuttered speech dataset and IIITH-IED dataset are used here to develop frame-level automatic disfluency detection systems for four types of disfluencies and the effect of SDC features on the detection of each disfluency type is examined using MFCC and SFFCC cepstral representations. Overall, it is found that using MFCC+SDC features gives an absolute improvement of 2.98% and 6.02% for stuttered and spontaneous speech disfluency detection respectively over the static MFCC features, while SFFCC+SDC features give an absolute improvement of 4.62% for stutter disfluencies and 7.28% for spontaneous speech disfluencies over the static SFFCC features, showing the importance of considering temporal variations for disfluency detection.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; *Speech recognition*; Artificial intelligence.

KEYWORDS

Speech Disfluencies, Single Frequency Filtering, Shifted Delta Cepstral Coefficients.

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1 INTRODUCTION

Disfluencies are the abrupt breaks or discontinuities in the normal speech flow. In speech, disfluencies may arise due to a number of reasons. In case of spontaneous or unprepared speech, messages have to be constructed on the fly [20]. The speaker might need to pause due to language complexity or to think about what has to be spoken next. Such scenarios lead to the presence of disfluencies. In case of stuttered speech, the speaker suffers from an impediment in which the speech flow is disrupted unintentionally by the presence of disfluencies [1]. Some of the most common types of disfluencies occurring in both these types of speech are -

- (1) Filled Pause - Fillers like ‘um’ and ‘uh’ which fill a speech pause but do not add meaning to the utterance.
- (2) Prolongation - Lengthening of any part of word, for example: lengthening of /o/ in ‘to’.
- (3) Word Repetition - Repeating a particular word to maintain the speech flow. For Example: I know you and... and him.
- (4) Part-Word Repetition - Repetition of a part of a word. For Example: y..you.
- (5) Phrase Repetition - Repeating a phrase of the utterance while taking time to think. For Example: I have...I have a book.

Detection of disfluencies is an important task for a number of applications. In case of stuttering, automatic detection of disfluencies helps in the clinical evaluation of stutter severity [4, 29] to decide the proper pathological treatment. For ASR systems, since most systems are developed for read and non-spontaneous speech, the presence of disfluencies lead to high word error rates (WER) [11, 31, 36]. Thus, the detection and removal of disfluencies is important for the efficient working of such systems.

Broadly speaking, disfluency detection has been explored in the literature in three ways - (1) using only lexical and text-based features extracted from the transcription [8, 37], (2) using only speech-based acoustic features [6, 34], (3) using a combination of both text-based and acoustic features [21, 22, 26, 32]. In this paper, we focus on the use of speech-based features for the task of disfluency detection. Disfluency detection directly from speech has been the focus of many studies [6, 9, 17]. In [30], local prosodic features like duration, fundamental frequency (F0) and signal-to-noise ratio (SNR) were used for the detection of four types of disfluencies. In [6, 35], the stability of formants and nasality effect (ratio of first two anti-formants) in the voiced regions of speech were used for

the detection of filled pause and vowel lengthening. Thresholding techniques based on energy features were explored in [14] for disfluency detection, and the effect of using different analysis window lengths was studied. In [28], log Mel-filterbank energies along with F0 contour was used for the frame-level detection of multiple disfluencies using SVM and DNN classifiers. MFCC features have been used extensively for the detection of speech disfluencies as well, and multiple modifications to the normal MFCC features have been explored in [12, 25, 27] to enhance the detection performance. Utterance-level disfluency detection was performed for six types of disfluencies in [19] using spectrogram as input to a Deep Residual network with Bidirectional LSTM (BiLSTM) network in 4-second audio files.

In order to study the effect of temporal variations of cepstral coefficients, shifted delta cepstral (SDC) coefficients have been used in [16, 18, 33]. In [18], SDC features were used for the dialect classification task, and significant improvements were reported. In the present work, SDC features have been used to effectively capture temporal variations in the speech signal due to disfluency occurrences. Four types of disfluencies- filled pause, prolongation, part-word repetition and word repetition from the UCLASS dataset for stuttered speech and IIITH-IED dataset have been considered to perform this study. Since the average duration of occurrences of each disfluency type is different, our aim here is to investigate the different configurations of SDC, which might help in improving the detection performance for the disfluencies in each of the datasets mentioned above. Furthermore, the use of Single Frequency Filtered Cepstral Coefficients (SFFCCs) [3, 13, 16], which provide high spectral resolution while computing the speech spectrum, has been investigated for the task of disfluency detection. The main contributions of this work are as follows:

- Investigating the use of SDC features to detect four types of speech disfluencies in both stuttered and spontaneous speech and explore different SDC configurations for each disfluency type.
- Investigating the use of SFFCC features for detecting disfluencies directly from the speech signal.

The rest of the paper is organized as follows - details about SDC feature extraction and SFFCC are discussed in Section 2. The disfluency detection system, the datasets used for building the system and evaluation metrics are described in Section 3. In Section 4, the details about the experiments performed and the results obtained are presented. Finally, we conclude by providing a summary of the current work and plans for future works in Section 5.

2 SHIFTED DELTA CEPSTRAL COEFFICIENTS AND SFFCC EXTRACTION

In this section, the extraction of Shifted Delta Cepstral (SDC) coefficients from the speech signal is discussed, followed by the extraction of Single Frequency Filtered Cepstral Coefficients (SFFCCs).

2.1 Shifted Delta Cepstral (SDC) Coefficients

SDC coefficients help in capturing temporal variations in the speech signal by stacking up delta cepstral features across several frames [18]. This serves as the motivation for investigating the effect of using SDC features for the disfluency detection task. The configuration

used for the computation of SDC features is governed by 4 parameters and is represented as $N - d - p - K$. Parameter N denotes the number of static cepstral coefficients computed per frame. For a speech frame t , the delta computations are performed between the cepstral coefficients of frames $(t + ip - d)$ and $(t + ip + d)$. So, the SDC coefficients for the t^{th} frame in the i^{th} iteration are obtained as follows:

$$\Delta c(t, i) = c(t + ip + d) - c(t + ip - d), \quad (1)$$

where the function $c(n)$ represents the cepstral coefficients for the n^{th} frame. The value of i is varied from 1 to K , giving K delta computations for the speech frame t . The final SDC features are obtained by stacking up all the delta coefficients obtained, giving the final $N * K$ dimensional SDC coefficients for frame t . The parameter d defines the delay or advance from the current frame, while the parameter p defines the time shift between successive delta computations. For the experiments performed in this work, SDC features are computed for MFCC and SFFCC cepstral coefficients.

2.2 Extraction of SFFCC

Single frequency filtering cepstral coefficients (SFFCCs) provide high spectro-temporal resolution for speech signals [3, 5]. In Single Frequency Filtering (SFF), the amplitude envelope for every frequency is computed as a function of time. This is done by multiplying the speech signal with an exponential function to frequency-shift the signal to the desired frequency. Let $s[n]$ be the speech signal. The frequency-shifted signal is given as:

$$\hat{s}[n] = s[n]e^{-j2\pi n\hat{f}_k/f_s} \quad (2)$$

where f_s is the sampling frequency and $\hat{f}_k = (f_s/2) - f_k$; f_k being the k^{th} desired frequency whose amplitude envelope has to be computed. $\hat{s}[n]$ is then filtered using a single pole filter with transfer function,

$$H(z) = \frac{1}{1 + az^{-1}} \quad (3)$$

$z = -a$ is the pole of the filter on the negative real axis. The filtered output obtained is:

$$y_k[n] = -ay_k[n-1] + \hat{s}[n] \quad (4)$$

The amplitude envelope of the filtered output is:

$$m_k[n] = \sqrt{(y_{kr}[n])^2 + (y_{ki}[n])^2} \quad (5)$$

Here, $y_{kr}[n]$ and $y_{ki}[n]$ are the real and imaginary parts of $y_k[n]$ respectively. The amplitude envelope computation can be done at multiple desired frequencies by defining f_k as:

$$f_k = k\Delta f, \quad k = 1, 2, \dots, \frac{(f_s/2)}{\Delta f} \quad (6)$$

where Δf is the interval at which desired frequencies are chosen. Using $m_k[n]$, the SFF spectrum can be calculated for any time instant. Finally, the SFFCC features are obtained by computing the cepstrum of the SFF spectrum:

$$S_k[n] = \text{IFFT}(\log_{10}(m_k[n])) \quad (7)$$

The first 13 coefficients of $S_k[n]$ are considered as the SFFCCs here. Figure 1 illustrates the steps involved in the computation of SFFCC.

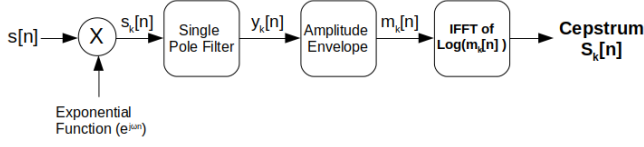


Figure 1: Extraction Process of Single Frequency Filtered Cepstral Coefficients

3 EXPERIMENTAL SETUP

In this section, we discuss the details about the UCLASS and IIITH-IED datasets, feature extraction process, the detection model and evaluation metrics used for comparing the results.

3.1 UCLASS Dataset

The UCLASS Dataset [15] has been used here to perform the disfluency detection experiments for stuttered speech. We have used the speech recordings from Release One of the UCLASS dataset, which consists of monologue recordings in British English from children of age 8 to 18 who suffer from stuttering. From the available recordings, 25 unique participants have been used here because of the availability of corresponding transcriptions for each monologue. The transcriptions are force aligned with the monologue recordings to get the timestamps for each word. The recordings have been annotated for multiple disfluencies. Here we use four disfluency types from this dataset - filled pause (interjection), prolongation, word repetition and part-word repetition (sound repetition samples also included in this class).

3.2 IIITH-IED Dataset

The IIITH-Indian English Disfluency (IIITH-IED) dataset was introduced in [10]. It consists of 10-hours of lecture-mode speech in Indian English. The lectures used to prepare this dataset belong to the following domains- Artificial Intelligence, Computer Science, Database Management etc. Ten-minute speech recordings from 60 speakers (30 male and 30 female) are annotated for different types of disfluencies at the word-level (identifying where the disfluency occurs in transcription) as well as the signal level (noting the start-time and end-time for every disfluency occurrence). Number of occurrences of the four types of disfluencies used in the experiments in this study and their average duration in this dataset are reported in Table 1.

Table 1: Number of occurrences of the four disfluency types in IIITH-IED Dataset and their average duration.

Disfluency Type	# of Occurences	Avg. Duration
Filled Pause	1428	0.395 sec
Prolongation	71	0.553 sec
Part-word Repetition	164	0.954 sec
Word Repetition	211	0.890 sec

3.3 Feature Extraction

MFCC features are used for developing the baseline systems here. A Hamming window of size 25 ms and a shift of 10 ms is used to

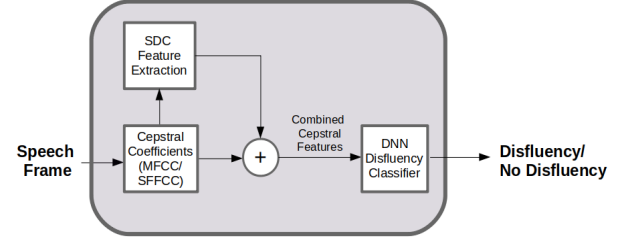


Figure 2: Disfluency detection pipeline used in this work.

extract 13-dimensional MFCC features per frame along with the energy of the frame. The 14-dimensional feature vector (13 MFCC+energy) obtained per frame is then mean-variance normalized and used to obtain SDC features. A modification is made while computing the SDC coefficients here. The value of i is varied from $-K$ to K while computing delta for every frame, rather than varying i from 1 to K as done in the standard SDC feature extraction process [18, 33]. Therefore, for every frame, a total of $N * (2K + 1)$ SDC coefficients are obtained. This is done so that delta computations are performed using frames occurring both before and after the current frame since the stability of the spectrum is a defining characteristic for disfluencies like filled pause and prolongation [6] (conventional SDC computation considers frames occurring after the current frame more for delta computations, see equation 1). For filled pause and prolongation, the SDC configuration used is 14-1-2-7, and 210 SDC coefficients are obtained, while for part-word and word repetition, 350 SDC coefficients are obtained using 14-1-2-12 configuration. The two configurations were chosen for these disfluencies based on their average duration in the datasets.

For the extraction of SFFCC features, the value of a is set to 0.985 while Δf is 20 Hz. The first 13 cepstral coefficients are extracted at an interval of 10 ms, giving the SFFCC features per frame.

3.4 Detection System

The disfluency detection systems used for the experiments are frame-level detection systems. These systems detect if a particular disfluency is present in a 10 ms speech frame or not, as done in [25, 28]. The detection of each type of disfluency is set up as a binary classification problem, and the same model architecture is used to detect each of the four types of disfluencies. The classifier used in the experiments is a DNN having 2 hidden layers. The number of hidden units in each hidden layer are 100 and 50, respectively. A train-test split of 80:20 is used in the experiments for each disfluency. While training, 10-fold stratified cross validation is performed, with 9 folds being used for training and 1 fold being used as the validation set. Grid search based hyperparameter optimization is performed for the DNN, and an optimal learning rate of 0.01 is used with the binary cross-entropy loss function and RMSprop optimizer. Early stopping is performed here on the detection loss of the validation set to avoid overfitting. The pipeline of the disfluency detection system is illustrated in Figure 2.

Both the IIITH-IED and UCLASS datasets are imbalanced since the number of occurrences of each disfluency are different. In order to ensure that the detection systems developed for each disfluency

are not biased, we use the Synthetic Minority Oversampling Technique (SMOTE) to ensure that the number of samples of each class are equal while performing the classification [7].

3.5 Evaluation metrics

Two evaluation metrics have been used - Detection accuracy and F1-score. F1-score gives a good idea of the system performance by balancing the false positives and false negatives, and hence, it has been used as an additional metric here.

4 RESULTS AND DISCUSSION

The detection experiments for each disfluency are performed for six types of feature configurations: (1) Static MFCCs (14-dimensional), (2) MFCC + conventional SDC, by varying i from 1 to K during delta computation (14-dimensional MFCC + 98-dimensional SDC features for 14-1-2-7 configuration for filled pause and prolongation; 168-dimensional SDC features using 14-1-2-12 configuration for part-word and word repetition), (3) MFCC + modified SDC, by varying i from $-K$ to K during delta computation (14-dimensional MFCC combined with 210 SDC features for filled pause and prolongation; 350 SDC features for part-word and word repetition), (4) Static SFFCCs (39-dimensional), (5) SFFCC + conventional SDC (K varied from 1 to 7 for filled pause and prolongation ; from 1 to 12 for part-word and word repetition in SDC computation), (6) SFFCC + modified SDC (K varied from -7 to 7 for filled pause and prolongation; from -12 to 12 for part-word and word repetition).

The detection results using the above mentioned feature configurations for each disfluency type in both the UCLASS and IIITH-IED datasets are presented in Table 2. From the table we can see that using SDC features along with the static cepstral coefficients gives better performance as compared to baseline MFCC/SFFCC systems for all four types of disfluencies. Further, using the modified version of SDC computation gave better detection accuracy and F1-score in both the datasets. In case of stuttered speech, an absolute improvement of 4.84% in the detection of filled pause using MFCC + modified SDC features and 6.11% using SFFCC + modified SDC features is obtained over the static cepstral features. This shows the importance of taking context from frames both before and after the current frame for delta computations in SDC since it would help in capturing temporal changes in the signal better. For prolongations in the UCLASS dataset, the improvements obtained are lesser as compared to filled pause (1.6% for MFCC + modified SDC features; 1.96% for SFFCC + modified SDC features). Improvements are obtained for the detection of repetition type disfluencies in the UCLASS dataset as well, with MFCC + modified SDC features giving an improvement of 2.93% and 2.53% for part-word and word repetitions respectively. The highest detection performance improvement is obtained in case of filled pause, since most instances of filled pause belong to the following two types - 'um' and 'uh' and the variations in the occurrences of the two types of filled pause can be captured effectively using SDC features. In case of the IIITH-IED dataset, the improvements obtained for each disfluency type are more as compared to the UCLASS dataset, since the number of samples for disfluencies are greater. The improvements obtained in the detection accuracy for filled pause and prolongation are 7.51%

and 6.81% respectively using MFCC along with modified SDC coefficients, while the improvement for repetition type disfluencies are 5.23% (part-word repetition) and 4.56% (word repetition). Averaging across all four disfluencies, we obtain an improvement of 2.98% for the stutter disfluencies, while an improvement of 6.02% is obtained for the spontaneous speech disfluencies using MFCC+ modified SDC features. In case of SFFCC, an average improvement of 4.62% is obtained for stutter disfluencies and 7.28% for spontaneous speech disfluencies using SFFCC + modified SDC features.

Another observation from the results in Table 2 is that, as compared to MFCCs, the performance of SFFCC features using the current disfluency detection system is less. This might be because, for SFFCC computation, the amplitude envelope is obtained for a single frequency, with an interval of $\Delta f = 20$ Hz used for envelope computation. These envelopes are then averaged to get the SFF spectrum. Whereas in case of MFCC, the frequency content of all frequencies is considered, which leads to a better cepstral representation in this case. Using SFFCC features for disfluency detection in the IIITH-IED dataset, the best performance is obtained with the SFFCC + modified SDC feature configuration, with the detection accuracy being 85.61% for filled pause, 87.07% for prolongations, 73.76% for part-word repetition and 78.93% for word repetitions.

Comparing the performance of our method to [19], which used acoustic features as well for disfluency detection in the UCLASS dataset, we find that our method gives better detection accuracy for filled pause, whereas for part-word repetitions, the accuracy is marginally lesser. In [23] [24], high accuracy is obtained for filled pause but a smaller subset of the UCLASS dataset is used in the experiments, preventing us from doing a fair comparison with our method. For prolongations and word repetitions however, [19] outperforms our method by a margin of 1.6% and 5.32% respectively (using MFCC + modified SDC configuration). This is because a fixed 4 second audio clip is used for spectrogram generation in [19], which makes it possible to capture disfluency region in prolongation and word repetition effectively. Compared to the detection methods which rely on statistical language models and task-based word lattices [2], our method gives better detection performance for sub-word disfluencies like part-word repetition, since methods like [2] depend heavily on the correctness of the orthographic transcription for detecting disfluencies.

From the results above, it is clear that using SDC features along with static cepstral coefficients (MFCCs/SFFCCs) gave improvements in the detection accuracies for the disfluency detection systems and produced high F1-scores. Using these observations, we further investigate different configurations of SDC features for the detection of each of the four disfluencies in both the datasets. Specifically, the important K parameter, which defines the number of delta computations to be performed for SDC (it also determines the length of the analysis window for delta computations), is varied to study the effect of the analysis window length on detection performance for every disfluency. For the experiments here, the MFCC + modified SDC features are used, since they give best detection performance for the four disfluencies. The other three parameters - N , d and p are kept fixed, with the value of N being 14 for MFCCs, while the values of d and p are 1 and 2, respectively.

The detection results obtained by varying the SDC configurations for filled pause and prolongation in the UCLASS dataset are reported

Table 2: Disfluency Detection results for four types of disfluencies in the UCLASS and IIITH-IED Datasets. The value of K for SDC features for filled pause and prolongation is 7 and for part-word and word repetitions $K=12$.

Dataset	Feature Configuration	Filled Pause		Prolongation		Part-Word Repetition		Word Repetition	
		Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score
UCLASS	MFCC	86.14	0.873	90.88	0.909	80.34	0.802	88.75	0.881
	MFCC+SDC (1 to K)	88.38	0.887	91.29	0.911	81.92	0.818	89.92	0.899
	MFCC+SDC ($-K$ to K)	90.98	0.910	92.48	0.927	83.27	0.831	91.28	0.911
	SFFCC	81.28	0.819	87.37	0.881	76.44	0.771	82.38	0.819
	SFFCC+SDC (1 to K)	84.33	0.850	89.16	0.892	78.29	0.787	86.75	0.867
	SFFCC+SDC ($-K$ to K)	87.39	0.879	89.33	0.895	80.93	0.811	88.31	0.880
IIITH-IED	MFCC	82.30	0.823	85.64	0.851	71.64	0.711	77.48	0.781
	MFCC + SDC (1 to K)	84.29	0.842	88.72	0.884	74.83	0.748	79.61	0.797
	MFCC + SDC ($-K$ to K)	89.81	0.901	92.45	0.926	76.87	0.772	82.04	0.819
	SFFCC	77.18	0.769	79.21	0.791	66.18	0.664	72.66	0.725
	SFFCC+SDC (1 to K)	81.43	0.815	83.77	0.839	71.85	0.719	75.71	0.757
	SFFCC+SDC($-K$ to K)	85.61	0.853	87.07	0.871	73.76	0.742	78.93	0.791

Table 3: Variation of parameter K in the SDC configuration for Filled Pause and Prolongation in UCLASS Dataset. Here, Acc. shows the Accuracy and F1 shows the F1-score.

Feature Configuration	Filled Pause		Prolongation	
	Acc.	F1	Acc.	F1
MFCC+SDC (14-1-2-5)	88.71	0.889	91.77	0.918
MFCC+SDC (14-1-2-7)	90.98	0.910	92.48	0.927
MFCC+SDC (14-1-2-9)	91.32	0.916	93.71	0.936
MFCC+SDC (14-1-2-11)	89.11	0.890	91.44	0.912

Table 4: Variation of parameter K in the SDC configuration for Filled Pause and Prolongation in IIITH-IED Dataset. Here, Acc. shows the Accuracy and F1 shows the F1-score.

Feature Configuration	Filled Pause		Prolongation	
	Acc.	F1	Acc.	F1
MFCC+SDC (14-1-2-5)	89.24	0.891	91.55	0.917
MFCC+SDC (14-1-2-7)	89.81	0.901	92.45	0.926
MFCC+SDC (14-1-2-9)	89.54	0.901	90.20	0.902
MFCC+SDC (14-1-2-11)	89.26	0.891	89.83	0.898

in Table 3 and for the IIITH-IED dataset the results are reported in Table 4. The value of K is varied from 5 to 11 with a step-size of 2 here. In case of the UCLASS dataset, the best detection results are obtained with the configuration 14-1-2-9. High detection accuracy of 91.32% and 93.71% are obtained for filled pause and prolongation. For the IIITH-IED dataset, marginal improvements in the detection performance of filled pause is observed by varying K . Here, the best performance is obtained with $K = 7$.

For the detection experiments of repetition type disfluencies the value of K is varied from 12 to 18 with a step-size of 2, since the average duration of repetition type disfluencies is more as compared to filled pause and prolongation. So a larger context window for delta computations in SDC has been considered. The results for repetition type disfluencies in the UCLASS dataset are presented in Table 5. As can be seen, the best detection results are obtained when the value of K is 14. When comparing the best performing

Table 5: Variation of parameter K in the SDC configuration for Part-word and Word repetition in UCLASS Dataset. Here, Acc. shows the Accuracy and F1 shows the F1-score.

Feature Configuration	PW Repetition		W Repetition	
	Acc.	F1	Acc.	F1
MFCC+SDC(14-1-2-12)	83.27	0.831	91.28	0.911
MFCC+SDC(14-1-2-14)	83.91	0.836	91.77	0.918
MFCC+SDC(14-1-2-16)	82.57	0.826	90.64	0.906
MFCC+SDC(14-1-2-18)	82.11	0.820	89.13	0.890

Table 6: Variation of parameter K in the SDC configuration for Part-word and Word repetition in IIITH-IED Dataset. Here, Acc. shows the Accuracy and F1 shows the F1-score.

Feature Configuration	PW Repetition		W Repetition	
	Acc.	F1	Acc.	F1
MFCC+SDC(14-1-2-12)	76.87	0.772	82.04	0.819
MFCC+SDC(14-1-2-14)	78.31	0.781	82.87	0.826
MFCC+SDC(14-1-2-16)	79.65	0.793	83.65	0.833
MFCC+SDC(14-1-2-18)	78.45	0.783	82.42	0.823

configuration ($K=14$) with the baseline MFCC results, an improvement of 3.57% is observed in the detection accuracy for part-word repetition, whereas for word repetition the observed improvement is 3.02%. The results for IIITH-IED dataset are reported in Table 6. Here, the best detection results are obtained with $K = 16$, with the accuracy for word repetitions being 83.65% and part-word repetitions being 79.65%. Compared to the baseline MFCC results on the IIITH-IED dataset, the best performing configuration ($K = 16$), the absolute improvement in the detection accuracy is 8.01% for part-word repetitions and 6.17% for word repetitions.

5 CONCLUSION

In this paper, the use of Shifted Delta Cepstral (SDC) coefficients was explored for the task of speech disfluency detection. The experiments were carried out for four types of disfluencies - filled pause, prolongation, part-word repetition and word repetition present

in the UCLASS stuttered speech dataset and the IIITH-IED spontaneous speech dataset. It was observed that using SDC features along with static cepstral coefficients (MFCCs/SFFCCs) produced significant improvements in the detection performance for all four disfluencies. We further explored a modified method for SDC computation to capture context from frames both before and after the current frame. With MFCC + modified SDC, an average improvement of 2.98% was obtained in the UCLASS dataset across all four disfluencies while the average improvement for IIITH-IED dataset was 6.02%. Future works will explore other acoustic features that can enhance the detection performance of individual disfluencies and use model-level adaptations to improve the disfluency detection pipeline further.

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